

# Global EV Market Analysis Project Report

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## 1. Project Overview and Objective

This project involves cleaning, transforming, and analyzing raw electric vehicle market data using Excel / Power Query and creating an interactive Power BI dashboard to derive meaningful business insights.

The main objective is to demonstrate data pre-processing techniques using Excel and an interactive Power BI dashboard visualization to track global vehicle electrification trends, monitor the expansion of public charging networks, and evaluate structural market shifts to make informed infrastructure decisions.

## 2. Data Sources

This project utilizes the public dataset from the **International Energy Agency (IEA)** Global EV Outlook.

Data Source : [https://www.iea.org/data-and-statistics/data-product/global-ev-outlook-2026\\*](https://www.iea.org/data-and-statistics/data-product/global-ev-outlook-2026*)

- Source Description and Timeline: Global EV market datasets compiled from public energy transition trackers and global infrastructure logs spanning a twenty-year timeline from 2010 through 2024, with forward-looking forecast points extending up to 2030.
- Domain: Automotive Analytics / Sustainability / Infrastructure Engineering / Sales Analytics

## 3. Objectives

- To analyze sales performance across regions and vehicle types to identify top-performing areas and adoption trends.
- To study charging infrastructure expansion patterns across countries and years for better capital allocation planning.
- To evaluate market penetration dynamics and structural changes for improved investment decision-making.
- To understand vehicle-to-charger ratios to enhance grid readiness and retail footprint optimization strategies.

#### 4. Attribute (Column / Features) Details

| Column Name                     | Data Type         | Description                                                                                                                                                                                    |
|---------------------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>region_country</b>           | String (Text)     | The primary geographic key column. Contains localized country records (e.g., India, USA, France, Germany) alongside broader continental groupings.                                             |
| <b>Region Type</b>              | String (Text)     | High-level geographical sorting classification used to partition data rows cleanly by geopolitical layout or operational tier.                                                                 |
| <b>category</b>                 | String (Text)     | The production reporting stream classification tier utilized for the row entry (e.g., Historical, Projections).                                                                                |
| <b>parameter</b>                | String (Text)     | The core industry metric being measured (e.g., EV stock share, EV sales, EV charging points, Battery demand, Electricity demand, Oil displacement).                                            |
| <b>mode</b>                     | Categorical       | The explicit transportation configuration segment being tracked (e.g., Cars, Vans, Trucks, Buses, 2 and 3 wheelers).                                                                           |
| <b>powertrain</b>               | Categorical       | The fuel technology type or commercial charging infrastructure hardware speed classification (e.g., BEV, PHEV, FCEV, Publicly available fast, Publicly available slow).                        |
| <b>year</b>                     | Numeric (Integer) | The chronological timeline tracking anchor or future projection period for the record, spanning from 2010 out to 2030.                                                                         |
| <b>value</b>                    | Numeric (Decimal) | The central numeric fact volume or percentage rate measured for the corresponding parameter combination.                                                                                       |
| <b>Market Category</b>          | String (Text)     | An analytical data filter engineered to flag aggregate rows (e.g., _World, EU27, Aggregate_sales_stock) vs. single regional elements (Countries Only) to prevent mathematical double-counting. |
| <b>Vehicle to Charger Ratio</b> | Numeric (Decimal) | Calculated key engineering data indicator tracking grid infrastructure load capacity.                                                                                                          |

## 5. Tools & Technologies

- Excel: Data cleaning, transformation, and Pivot Tables.
- Power BI: Data modelling, DAX calculations, visualization custom tooltip architecture, and interactive dashboard creation.

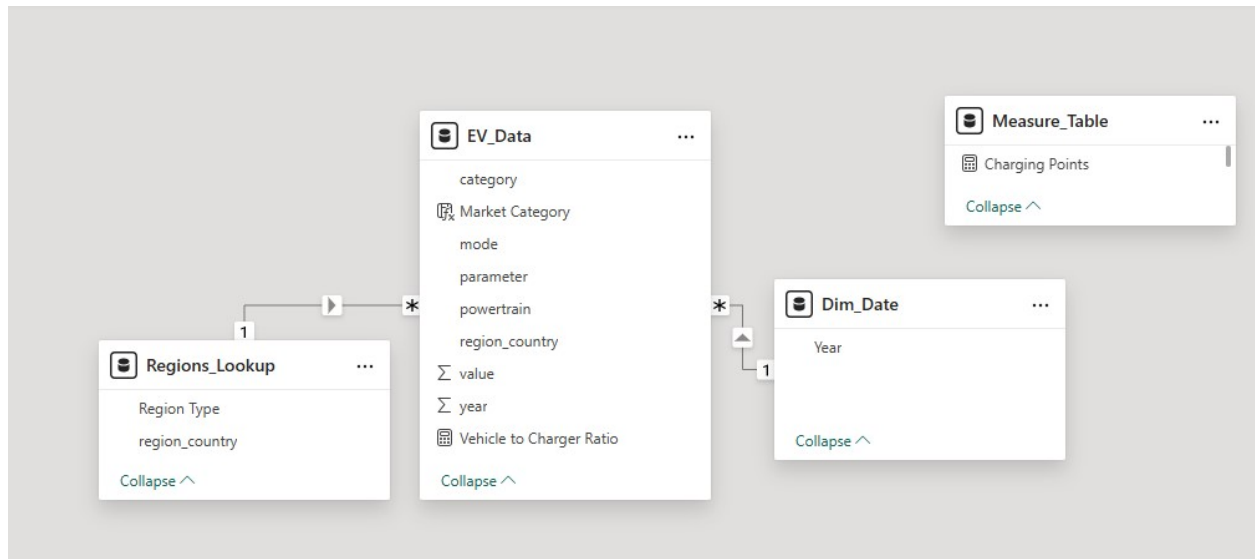
## 6. Data Pre-Processing (Excel / Power Query)

- Data Ingestion: Loaded the raw global electric vehicle dataset containing over 16,000 records spanning from 2010 to 2024, with projection data up to 2030.
- Row-Level Filtering: Cleaned and filtered rows under the category column to isolate core tracking metrics, and removed unneeded scenario projection markers such as Projection-STEPS.
- Column Value Standardization: Standardized attributes within the parameter and mode columns to ensure consistent category spelling (e.g., separating values like Cars, Vans, Trucks, Buses, and 2 and 3 wheelers).
- Powertrain Categorization: Filtered and cleaned the powertrain field to isolate plug types and technology classifications cleanly (BEV, PHEV, FCEV, Publicly available fast, and Publicly available slow).
- Data Type Configuration: Set explicit data types across the entire schema table (e.g., establishing year as an integer, value as a decimal number, and region\_country as text strings) to guarantee accurate dashboard visual execution.

## 7. Data Modelling and DAX (Power BI)

Data Model: Implemented a professional Star Schema relational model built on 1-to-many (1:\*) relationships with clean, single-direction cross-filtering to guarantee optimal visual rendering performance:

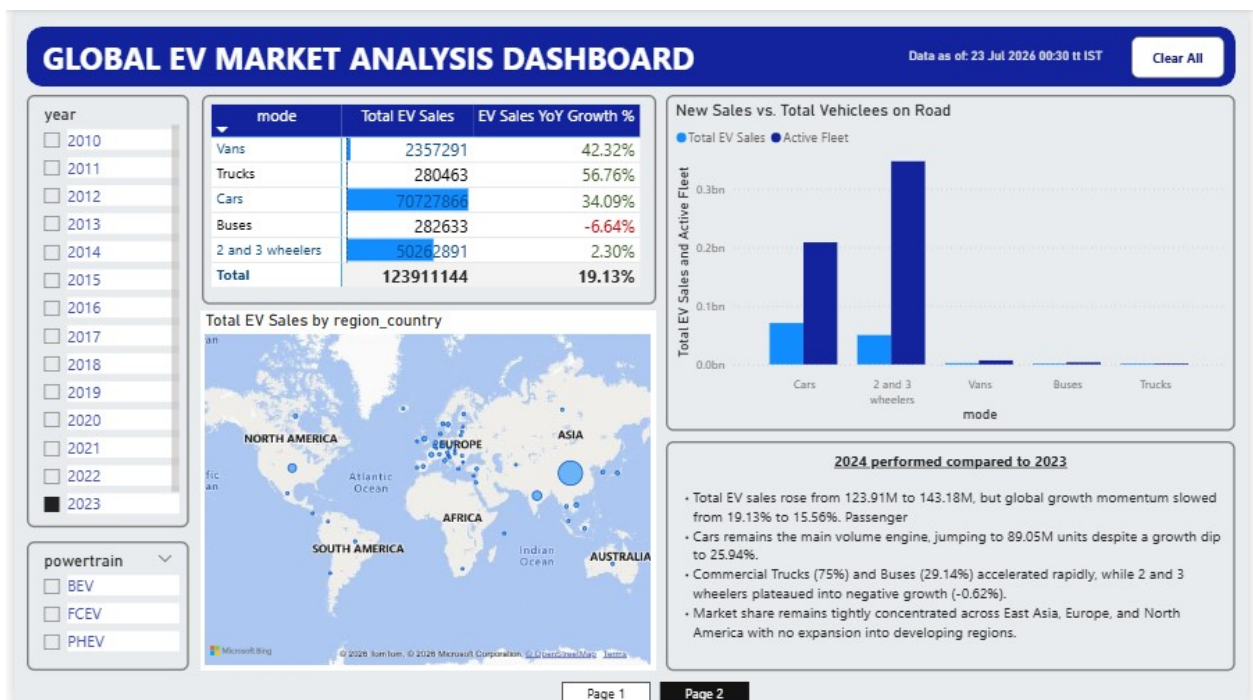
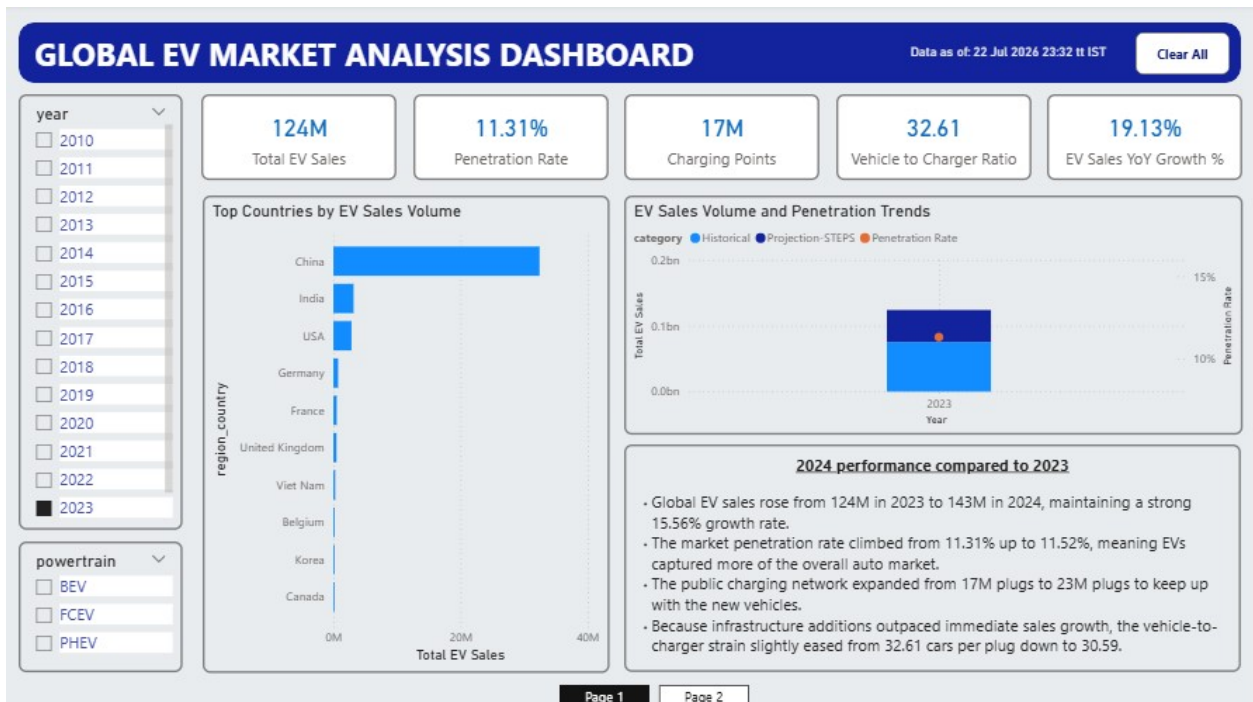
- **EV\_Data (Fact Table):** The central transactional repository housing core market values, containing structural parameters for category, market category, mode, parameter, powertrain, year, value, and the calculated metric field Vehicle to Charger Ratio.
- **Regions\_Lookup (Dimension Table):** Filters the fact table via a 1-to-many (1:\*) relationship mapped directly on the region\_country key column. It provides geographic grouping logic through the Region Type column.
- **Dim\_Date (Dimension Table):** Serves as the master time intelligence calendar table, filtering the fact table via a 1-to-many (1:\*) relationship mapped directly on the Year column.



### DAX Calculations:

- **Charging Points**: Aggregates public and private plug deployment trends.
- **EV Sales YoY Growth %**: Computes year-over-year adoption velocity deltas.
- **Penetration Rate**: Calculates EV adoption relative to broad market automotive options.
- **Total EV Sales**: Computes running transaction volumes.
- **Total EV Stock**: Aggregates the standing active vehicle fleet count.
- **Total Volume**: Baseline analytical counter for underlying record verification.
- **Vehicle to Charger Ratio**: Computes infrastructure capacity metrics to evaluate network strain.
- **Last Refreshed Text**: Generates automated report-wide UTC time stamping.

## 8. Analysis and Visualizations (Power BI)



**Dashboard Features:**

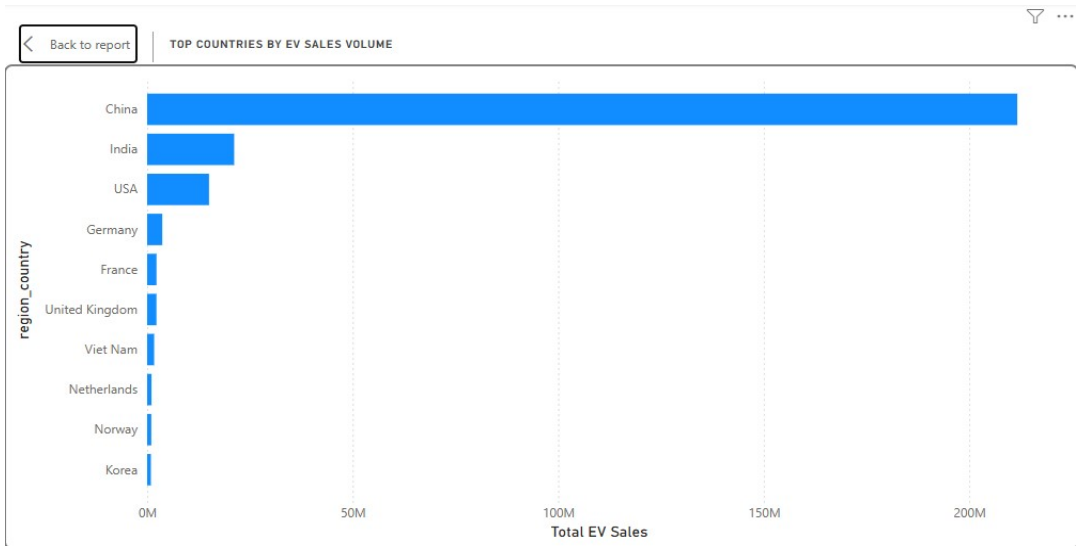
- KPI Summary Cards: High-level analytical visual indicators capturing Total EV Sales (143M), Penetration Rate (11.52%), Charging Points (23M), Vehicle to Charger Ratio (30.59), and EV Sales YoY Growth % (15.56%).



- Matrix Table Visual: Tracks detailed breakdowns of vehicle modes against individual sales volumes and explicit YoY percentage movement metrics.
- Clustered Bar Chart: Compares horizontal performance variables between new sales segments and active vehicle fleet volumes on the road.
- Custom Hover Tooltip: A specialized pop-up layout linked to the bar chart that maps regional data insights dynamically (e.g., breaking down top contributors like China at 123M, Asia Pacific at 76M, India at 20M, Europe at 3M, Viet Nam at 2M, and EU27 at 1M).
- Clear Titles: All Filters reset button to optimize navigation.

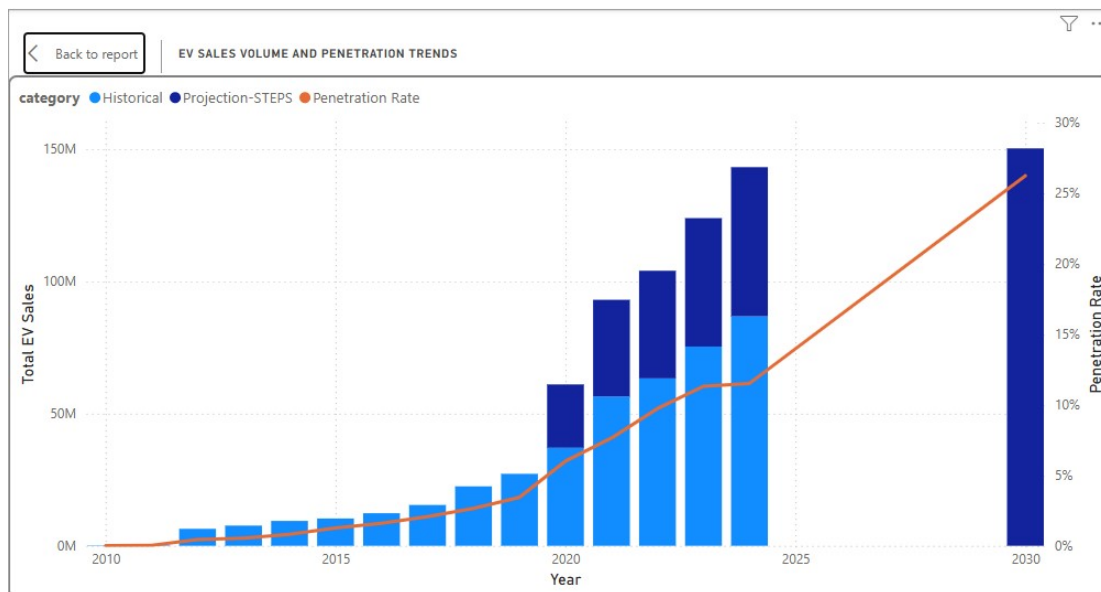
## Top Countries by EV Sales Volume

- Compares and ranks individual nations based on their absolute electric vehicle transaction volumes.
- Key details: Visually isolates market leaders, showing China dominating global adoption by an overwhelming margin over the USA and Germany.
- Business value: Tells stakeholders exactly which geographic territories hold the highest consumer demand and manufacturer density.



## EV Sales Volume and Penetration Trends (Line and Stacked Column Chart)

- It combines two different metrics on a single chronological timeline to track how absolute market volumes (Total EV Sales) grow alongside a relative industry benchmark (Penetration Rate) from 2010 out to 2030.
- Business value: It allows planners to simultaneously monitor the past scaling path and future market depth on a unified horizon, making it easy to see if market penetration rates are safely keeping pace with structural sales volumes.



## Vehicle Segment Performance Matrix (Table)

- Breaks down sales volumes and growth velocity by specific vehicle transit forms (mode).
- Key details: Shows Cars as the highest volume driver (\$89.05\text{M}\$ units), Vans as the fastest expanding sector (\$42.32\%\$), and a slight market saturation drop in 2 and 3 wheelers (\$-0.62\%\$).
- Business value: Uses automated conditional formatting to highlight negative values in soft red, letting managers spot underperforming categories instantly.

| mode             | Total EV Sales   | EV Sales YoY Growth % |
|------------------|------------------|-----------------------|
| Vans             | 2357291          | 42.32%                |
| Trucks           | 280463           | 56.76%                |
| Cars             | 70727866         | 34.09%                |
| Buses            | 282633           | -6.64%                |
| 2 and 3 wheelers | 50262891         | 2.30%                 |
| <b>Total</b>     | <b>123911144</b> | <b>19.13%</b>         |

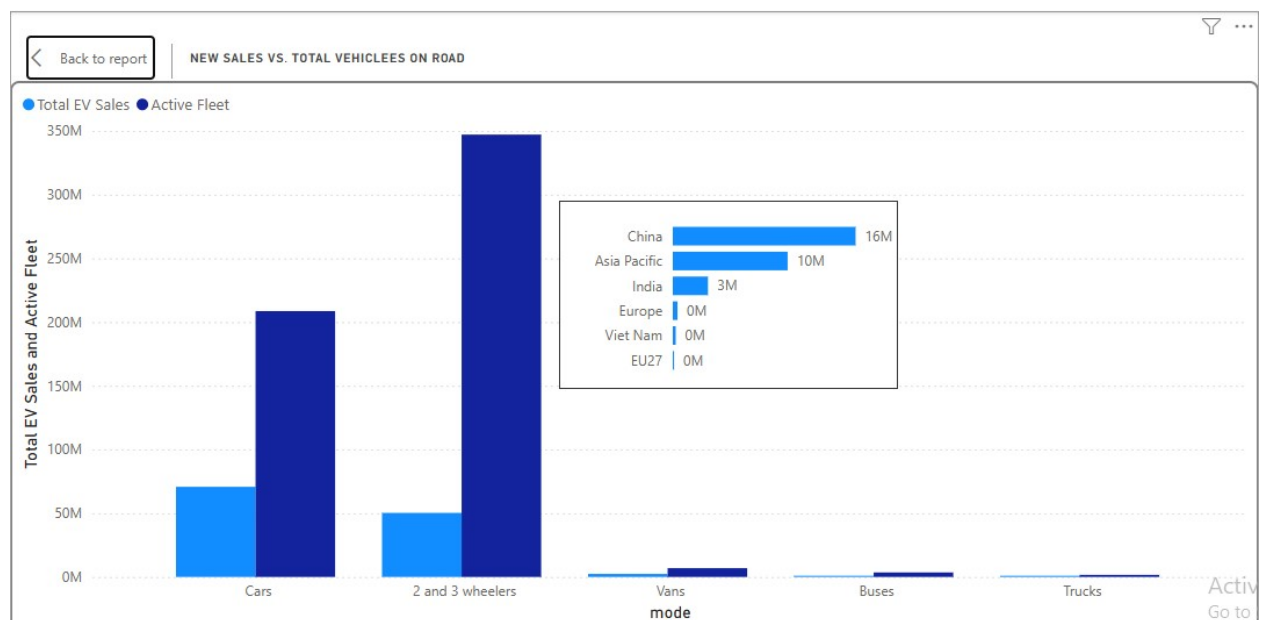


## Total EV Sales and Active Fleet (Clustered Bar Chart)

- Directly compares newly added vehicle sales volumes against the total cumulative operational vehicles remaining on the road.
- Key details: Maps data side-by-side across all transportation modes to show total scaling over time.
- Business value: Helps infrastructure engineers gauge long-term grid demand by measuring active rolling stock rather than just short-term sales cycles.

## Interactive Custom Hover Tooltip

- Provides a hidden, on-demand micro-chart detailing regional distribution.
- Key details: Automatically pops up when hovering over any bar visual to display localized
- Numbers Business value: Maximizes valuable screen canvas layout space by hiding complex geographical breakdowns until the user actively seeks them.



## 9. Analysis Insights

- **Descriptive:** Global vehicle penetration rates reached an all-time baseline maximum of 11.52%, driven heavily by the rapid commercial scaling of battery-electric passenger configurations.
- **Diagnosis:** While heavy trucks and commercial transport segments showed rapid acceleration spikes, the 2 and 3 wheeler mobility tier plateaued into a slight negative growth delta of -0.62%, indicating local structural market saturation or changes to regional subsidy programs.
- **Predictive:** Based on historical baseline slopes, infrastructure additions must sustain a compounding annual trajectory above 18% to prevent regional grid overloads as passenger adoption scales toward 2030.
- **Prescriptive:** Investment strategies should aggressively redirect grid capital away from saturated premium passenger zones and shift focus toward developing fast-charging freight corridors to support commercial truck scaling.

## 10. Conclusions

The integration of Excel, Power Query, and Power BI proved highly effective for end-to-end data analysis. Transforming a raw dataset of over 16,000 messy records into a clean, star-schema relational model successfully streamlined data processing. By layering on advanced features like synchronized slicers, conditional matrix highlighting, and custom pop-up hover tooltips, the final dashboard serves as a production-grade asset that delivers rapid, scannable, and actionable market intelligence for global stakeholders.